

Criteo Uplift Analytics Engine

An Advanced Machine Learning decision engine for optimising ad spend, identifying persuadable audiences, maximising ROI

Enter Dashboard

Conversion Value

29.00

Cost per Ad

\$0.01

Scale Factor

\$1,000,000

\$0.08

Net Profit - AB Strategy

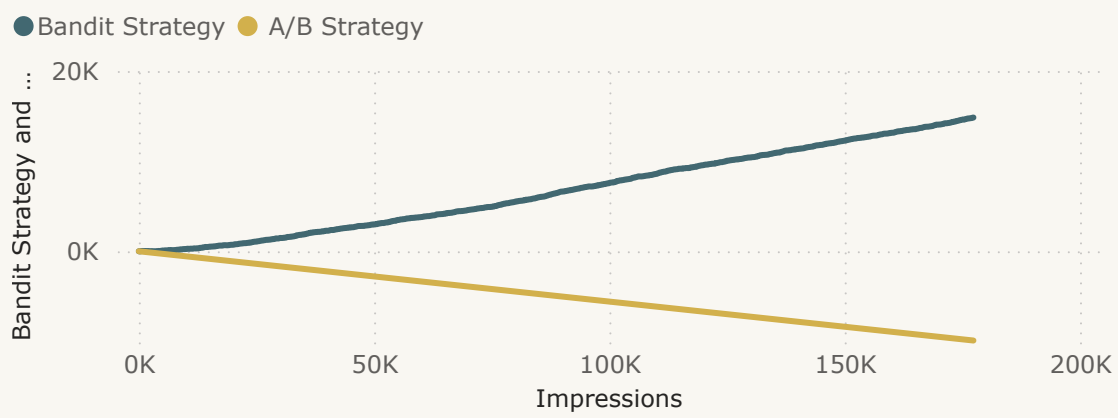
\$0.08

Net Profit - Bandit Strategy

\$0.01

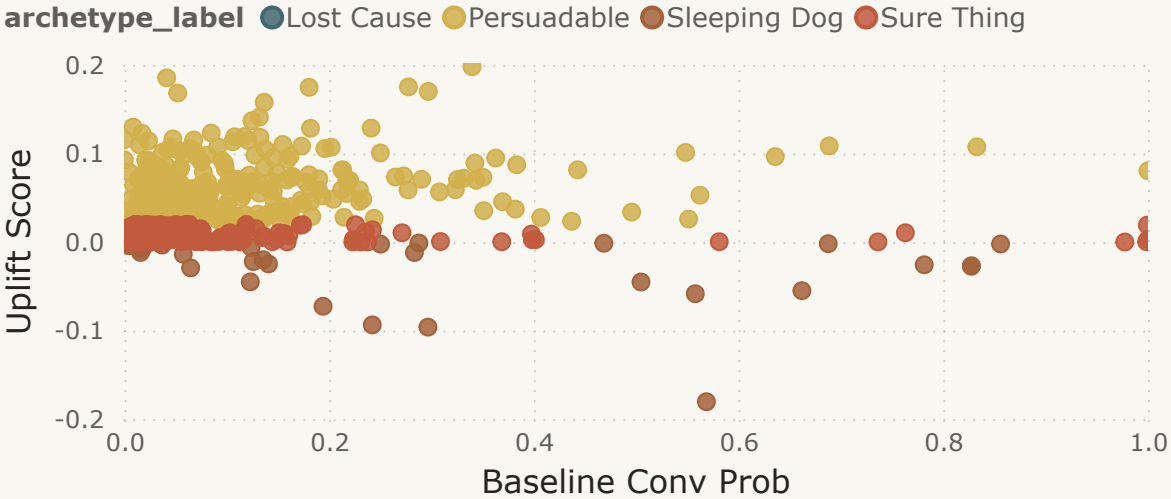
Profit Delta

Bandit vs. A/B Trajectory

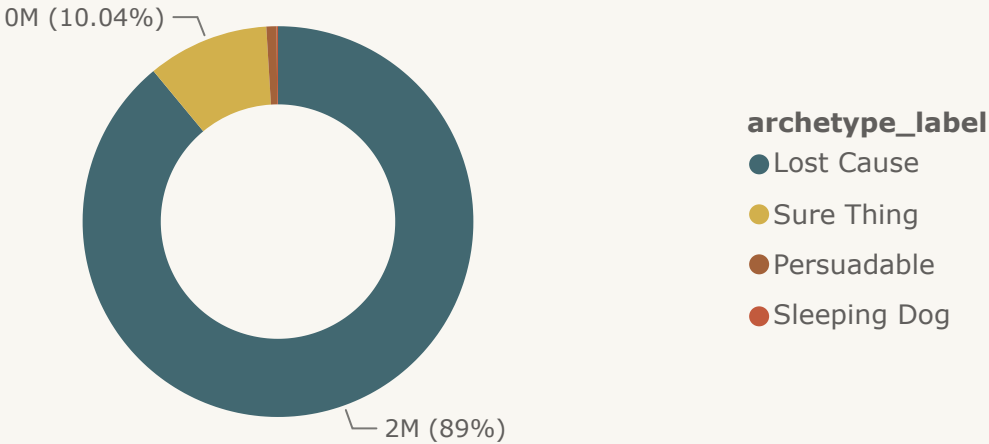


Strategy	Profit	Total Regret
A/B Random (Baseline)	-0.06	10000
LinUCB Bandit (Optimal)	0.08	500
Uplift Model (Greedy)	0.08	2000

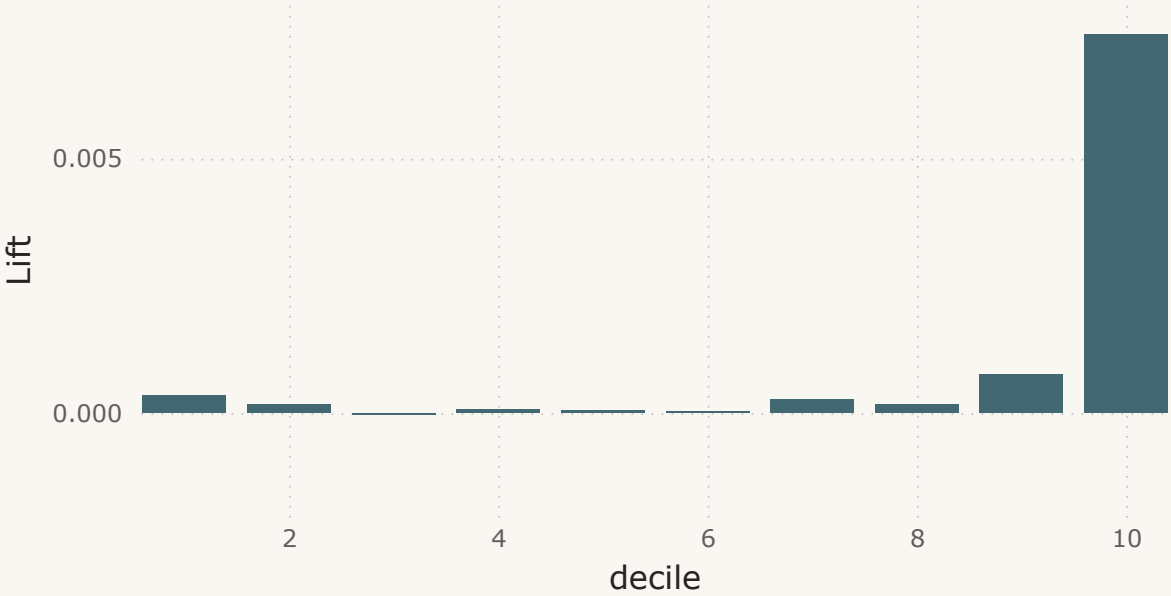
Archetype Clustering



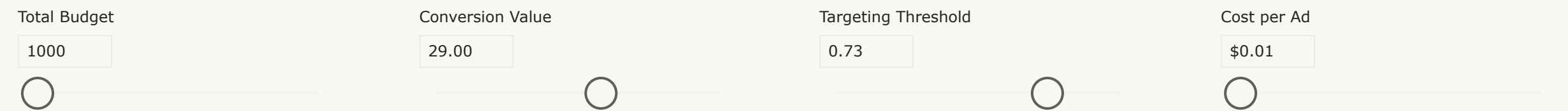
Users by Archetype



Lift by decile



Feature	Persuadables	Others	Diff %
f0	16.14	20.24	-20.24
f1	10.12	10.06	0.53
f10	5.52	5.30	4.10
f11	-0.18	-0.17	-8.32
f2	8.48	8.44	0.44
f3	2.50	4.48	-44.15
f4	10.65	10.28	3.60
f5	3.78	4.07	-7.22
f6	-7.43	-3.58	-107.76
f7	5.65	5.01	12.81
f8	3.84	3.95	-2.79
f9	28.21	13.87	103.38



10.21M

Projected Audience Size

\$102.05K

Projected Cost Total

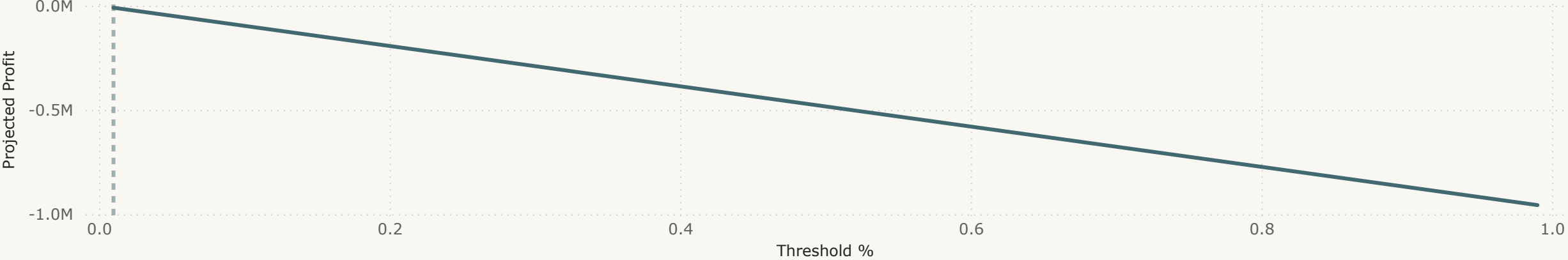
33K

Projected Conversions

\$858.05K

Projected Net Profit Total

Profit Diminishing Returns



HEALTHY

Experiment Health Score

PASS

SRM Status

PASS

Covariate Balance Status

2M

Control Users

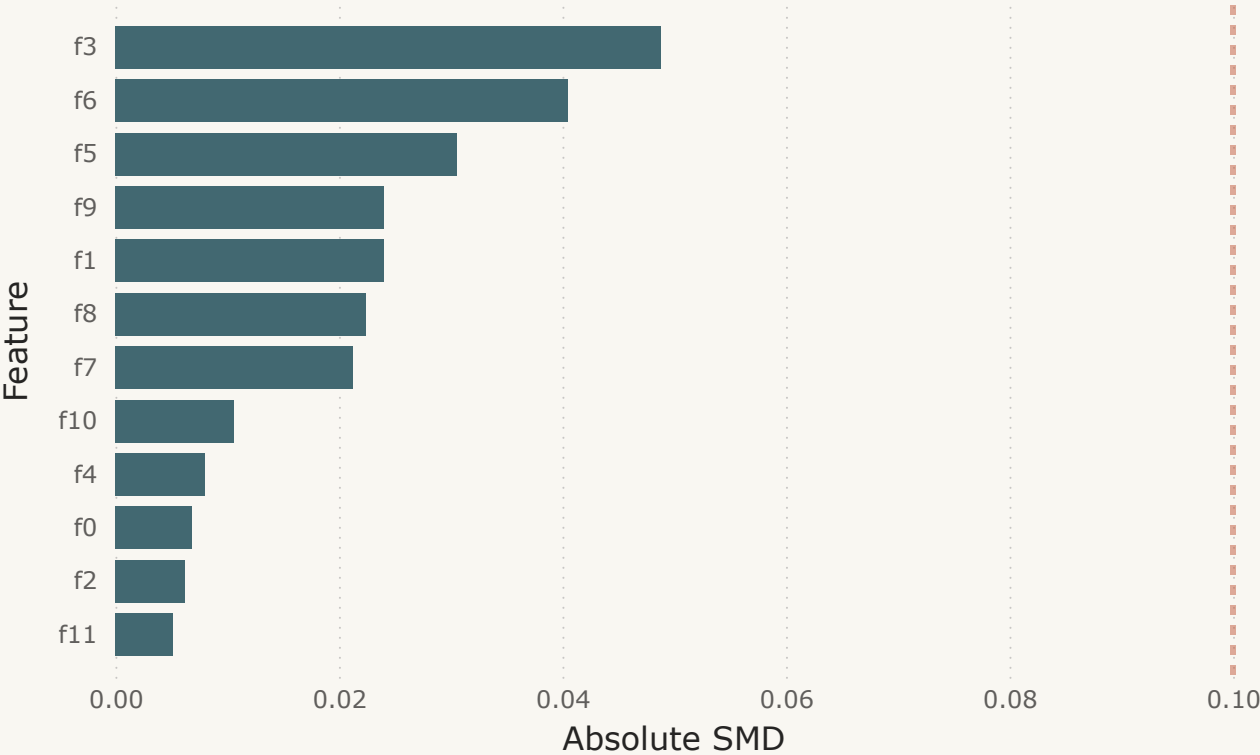
12M

Treatment Users

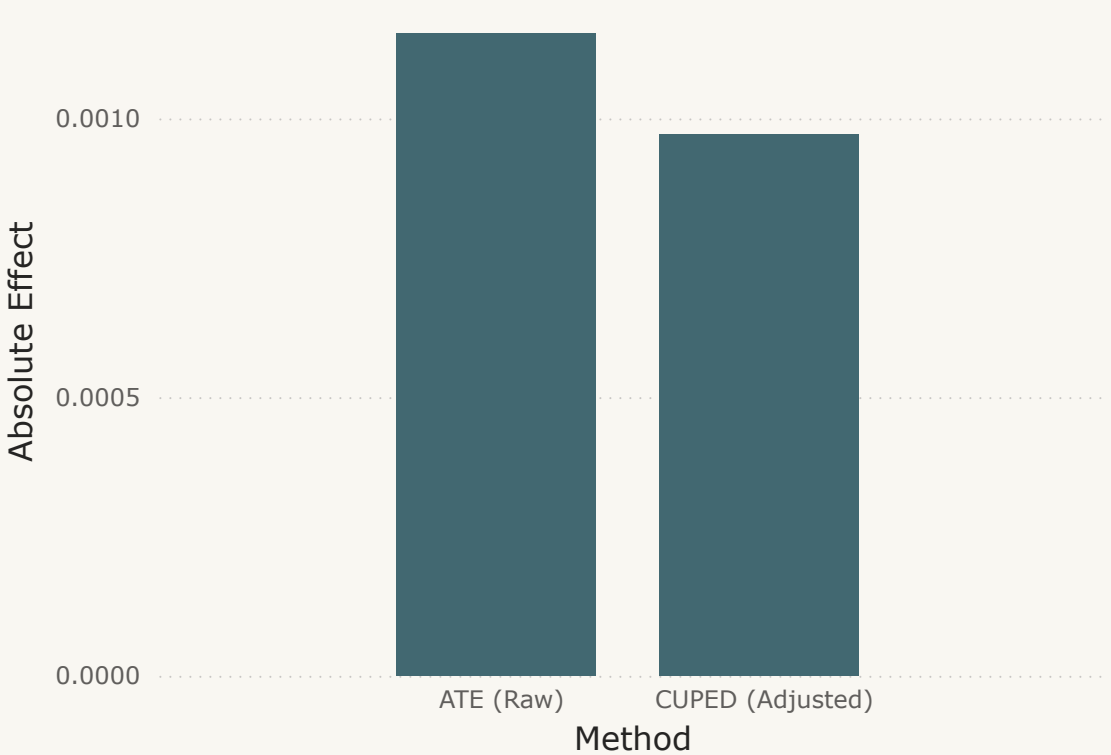
14M

Total Users

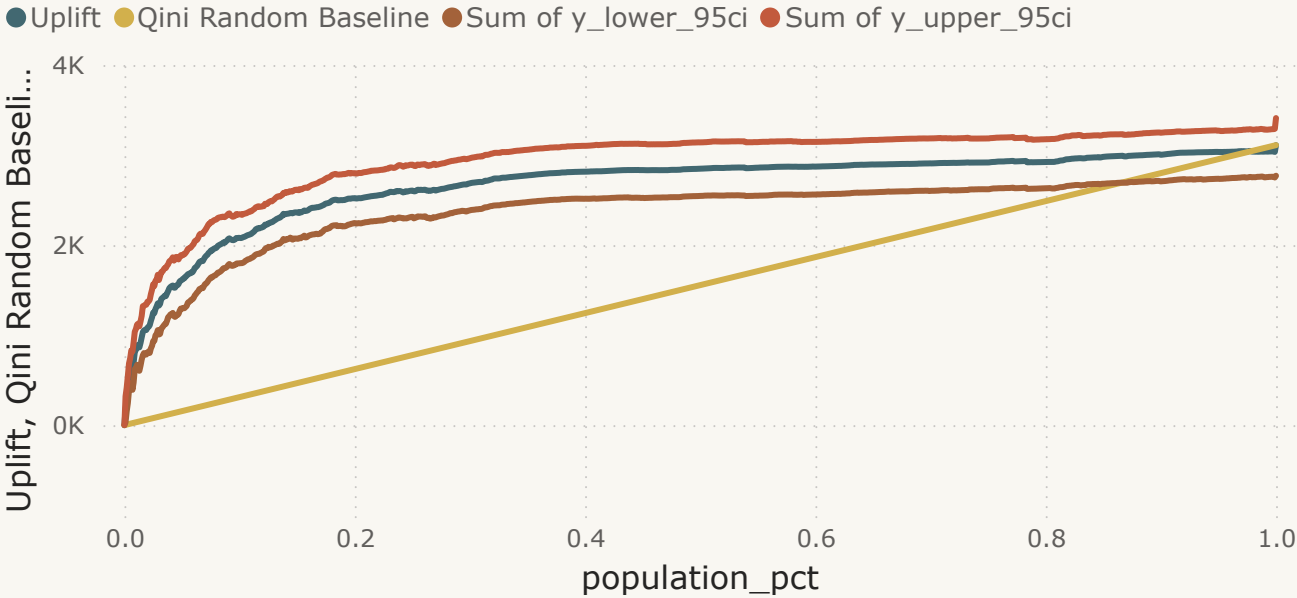
Covariate Balance (SMD)



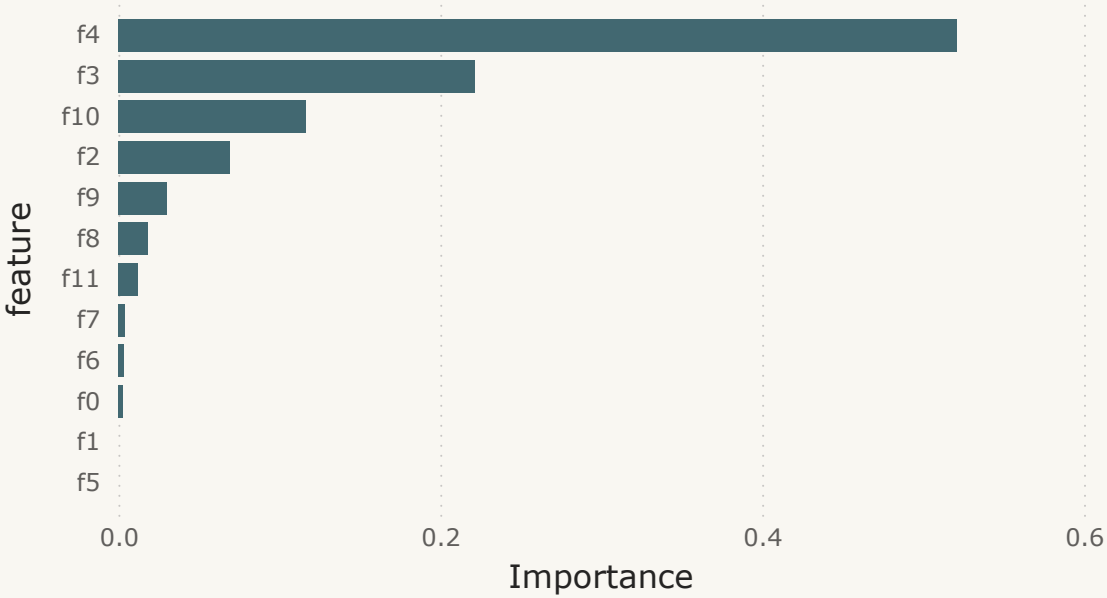
Absolute Effect by Method



Qini Curve (Model Performance)



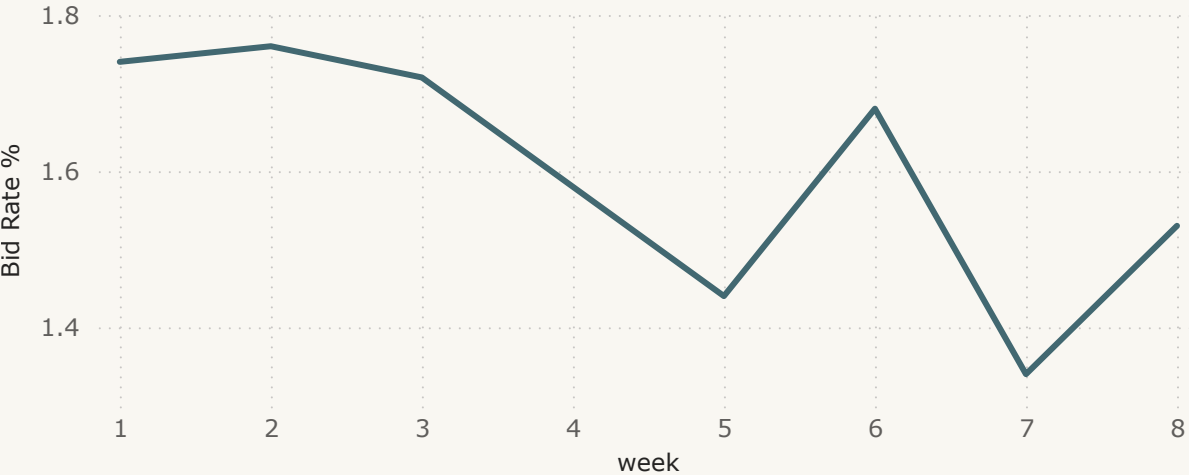
Importance by feature



archetype_hint	Depth	Feature	n_samples	node_id	node_type	predicted_uplift	rule_path	threshold
Low Uplift	5	—	135	61	Leaf	0.00	→ f4 > 12.14 → f3 > 3.15 → f10 > 5.57 → f4 > 15.49 → f9 ≤ 41.08	—
Low Uplift	5	—	4693	59	Leaf	0.02	→ f4 > 12.14 → f3 > 3.15 → f10 > 5.57 → f4 ≤ 15.49 → f9 > 26.28	—
Low Uplift	5	—	4373	58	Leaf	0.01	→ f4 > 12.14 → f3 > 3.15 → f10 > 5.57 → f4 ≤ 15.49 → f9 ≤ 26.28	—
Low Uplift	5	—	414	43	Leaf	0.01	→ f4 > 12.14 → f3 ≤ 3.15 → f2 > 8.39 → f9 ≤ 26.28 → f4 ≤ 12.45	—
Low Uplift	5	—	64859	31	Leaf	0.00	→ f4 ≤ 12.14 → f10 > 5.30 → f3 > 2.17 → f11 > -0.30 → f3 > 3.15	—
Low Uplift	5	—	6398	30	Leaf	0.01	→ f4 ≤ 12.14 → f10 > 5.30 → f3 > 2.17 → f11 > -0.30 → f3 ≤ 3.15	—
Low Uplift	5	—	1912	28	Leaf	0.01	→ f4 ≤ 12.14 → f10 > 5.30 → f3 > 2.17 → f11 ≤ -0.30 → f10 > 5.57	—
Low Uplift	5	—	2013	23	Leaf	0.01	→ f4 ≤ 12.14 → f10 > 5.30 → f3 ≤ 2.17 → f4 > 11.30 → f9 ≤ 26.28	—
Low Uplift	5	—	9455	21	Leaf	0.01	→ f4 ≤ 12.14 → f10 > 5.30 → f3 ≤ 2.17 → f4 ≤ 11.30 → f8 > 3.80	—
Low Uplift	5	—	126074	16	Leaf	0.00	→ f4 ≤ 12.14 → f10 ≤ 5.30 → f8 > 3.80 → f9 > 24.85 → f6 > -9.94	—
Low Uplift	5	—	31631	15	Leaf	0.00	→ f4 ≤ 12.14 → f10 ≤ 5.30 → f8 > 3.80 → f9 > 24.85 → f6 ≤ -9.94	—
Low Uplift	5	—	2367196	13	Leaf	0.00	→ f4 ≤ 12.14 → f10 ≤ 5.30 → f8 > 3.80 → f9 ≤ 24.85 → f3 > 0.91	—

feature	1	2	3	4	5	6	7	8
f0	0.21	0.20	0.21	0.04	0.38	0.23	0.13	0.21
f1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
f10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
f11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
f2	0.16	0.16	0.16	0.02	0.24	0.15	0.10	0.00
f3	0.03	0.03	0.03	0.02	0.00	0.05	0.00	0.08
f4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
f5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
f6	0.14	0.14	0.14	0.01	0.22	0.11	0.08	0.01
f7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
f8	0.22	0.23	0.23	0.03	0.36	0.26	0.19	0.01
f9	0.09	0.10	0.09	0.02	0.07	0.10	0.06	0.00

Bid Rate Trend



Expected Profit Trend

